

Formula

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A^c) = 1 - P(A)$; $P(A|B) = \frac{P(A \cap B)}{P(B)}$
2. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$
3. For n events $A_1, A_2, \dots, A_m, \dots, A_n$; $P(A_m|B) = \frac{P(B|A_m)P(A_m)}{P(B)} = \frac{P(B|A_m)P(A_m)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$
4. $F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$; $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y)dydx$
5. $f(x) = \frac{d}{dx}F(x)$; $f(a \leq x \leq b) = \int_a^b f(x)dx$
6. $f(x, y) = \frac{d^2}{dxdy}F(x, y)$; $f(x) = \int_y f(x, y)dy$
7. $f_{X|Y}(x > a|y < b) = \frac{f(x > a, y < b)}{f(y < b)}$; $f(x > a, y < b) = \int_{x=a}^{\infty} \int_{y=-\infty}^b f(x, y)dydx$
8. $E(X) = \begin{cases} \sum x_i P(x_i); & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x)dx; & \text{if } X \text{ is continuous} \end{cases}$; $E(X^2) = \begin{cases} \sum x_i^2 P(x_i); & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^2 f(x)dx; & \text{if } X \text{ is continuous} \end{cases}$
9. $V(X) = E(X^2) - [E(X)]^2$; $CV(X) = \frac{SD(X)}{E(X)}$; $SD(X) = \sqrt{V(X)}$
10. $E(aX + b) = aE(X) + b$; $V(aX + b) = a^2V(X)$; where a and b are constants
11. $f(x) = \binom{n}{x} p^x q^{n-x}$; $x = 0, 1, \dots, n$, $0 \leq p \leq 1$, $q = 1 - p$
12. $f(x) = pq^x$; $x = 0, 1, 2, \dots$, $0 \leq p \leq 1$, $q = 1 - p$
13. $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$; $x = 0, 1, 2, \dots$, $\lambda > 0$
14. $f(x) = \lambda e^{-\lambda x}$; $x > 0$, $\lambda > 0$
15. $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$; $-\infty \leq x \leq \infty$, $-\infty \leq \mu \leq \infty$, $\sigma^2 \geq 0$
16. $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$; $-\infty \leq z \leq \infty$, where $Z = \frac{X - \mu}{\sigma}$ and $X \sim N(\mu, \sigma^2)$
17. $P(a \leq X \leq b) = F(b) - F(a)$; $F(a) = P(X \leq a)$; $P(X > a) = 1 - P(X \leq a)$
18. $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$; $T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$; $s = \sqrt{\frac{\sum x_i^2 - n\bar{x}^2}{n-1}}$